

Question:

Melanie is a door-to-door saleswoman. She sold a third of her vacuum cleaners at the green house, 2 more to the red house, and half of what was left at the orange house. If Melanie has 5 vacuum cleaners left, how many did she start with?

Reasoning + Answer:

<think>

let's denote the initial number of vacuum cleaners Melanie had as x

at the green house , she sold a third of x , so she sold $x / 3$ vacuum cleaners

she was left with $x - x / 3 = 2 x / 3$ vacuum cleaners

at the red house , she sold 2 more , so she was left with $2 x / 3 - 2$

at the orange house , she sold half of what was left , so she sold $(2 x / 3 - 2) / 2 = x / 3 - 1$ vacuum cleaners

the number of vacuum cleaners she had left after the orange house is $2 x / 3 - 2 - (x / 3 - 1) = x / 3 - 1$

we are given that she has 5 vacuum cleaners left , so $x / 3 - 1 = 5$

$x / 3 = 6$

$x = 18$

</think>< answer > 18 </answer>

